

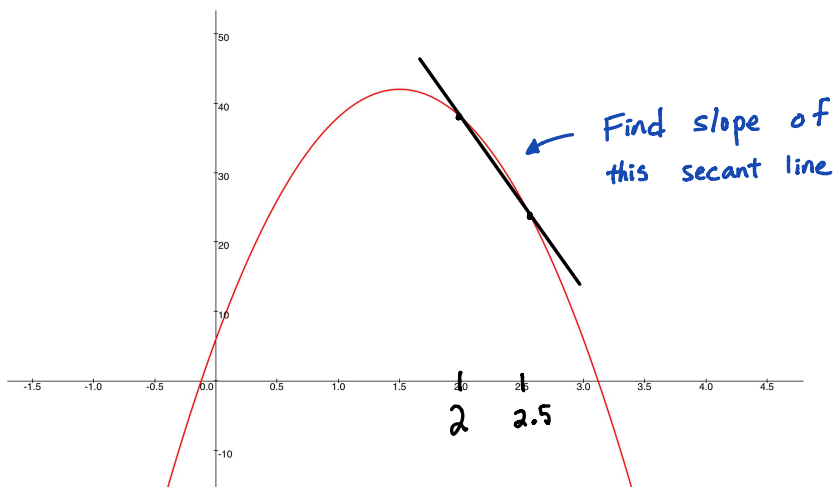
Section 2.1 Tangents And Velocities

Ex ① Given $f(x) = x^2$, evaluate $\frac{f(x) - f(3)}{x - 3}$ $\frac{\Delta y}{\Delta x} = \text{slope}$

$$\frac{f(x) - f(3)}{x - 3} = \frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x-3} = \boxed{x+3}$$

Ex ② The height of a ball thrown into the air is $h(t) = -16t^2 + 48t + 6$ feet after t seconds.

A) Find the Average velocity over $[2, 2.5]$.



$$h(2.5) = -16(2.5)^2 + 48(2.5) + 6 = 26 \text{ ft.}$$

$$h(2) = -16(2)^2 + 48(2) + 6 = 38 \text{ ft.}$$

$$\text{Average velocity} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{26 - 38}{2.5 - 2} \frac{\text{ft}}{\text{sec}}$$

It is
going down

$$\Rightarrow -24 \frac{\text{ft}}{\text{sec}}$$

B) What is the instantaneous velocity
when $t = 2$ seconds?

First use the interval $[2, 2.1]$

$$\text{Average velocity} = \frac{h(2.1) - h(2)}{2.1 - 2}$$

$$= \frac{36.24 - 38}{0.1}$$

$$= -17.6 \frac{\text{ft}}{\text{sec}}$$

Now use the interval $[2, 2.01]$

$$\text{Average velocity} = \frac{h(2.01) - h(2)}{2.01 - 2}$$

$$= \frac{37.838 - 38}{0.01}$$

$$= -16.2 \frac{\text{ft}}{\text{sec}}$$

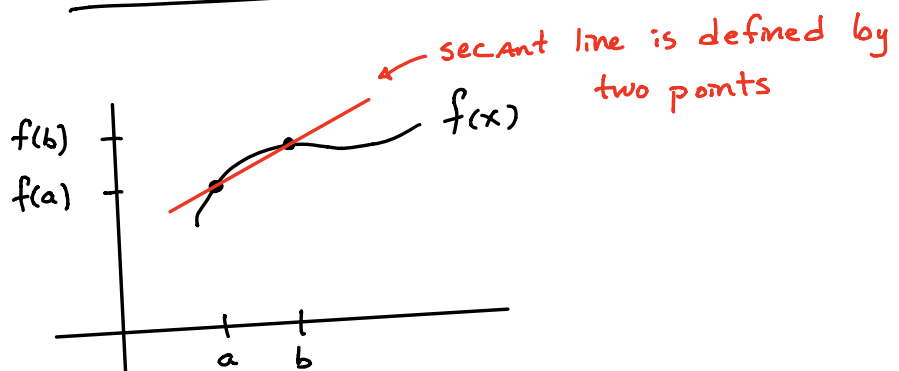
Now use the interval $[2, 2.001]$

$$\text{Average velocity} = \frac{h(2.001) - h(2)}{2.001 - 2}$$

$$= \frac{37.984 - 38}{0.001}$$

$$= -16 \frac{\text{ft}}{\text{sec}}$$

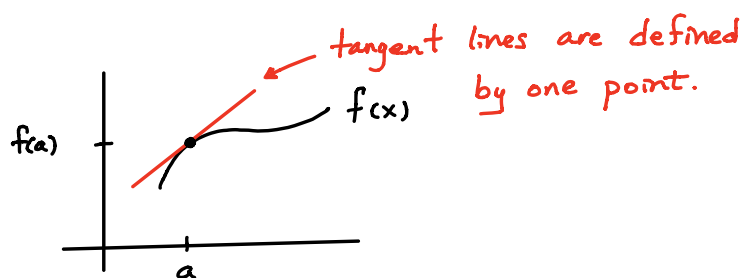
Secant Lines



$$\text{Slope} = \frac{f(b) - f(a)}{b - a} \quad \frac{\Delta y}{\Delta x}$$

Slope of the secant line gives us the Average velocity over the interval $[a, b]$.

Tangent Lines



Slope of the tangent line gives us the instantaneous velocity at the time a .

Ex (2) $f(x) = \frac{1}{1+x^2}$

Find the equation of the secant line over the interval $[0, 1]$.

$$\text{Slope} = \frac{f(1) - f(0)}{1 - 0}$$

$$= \frac{1}{1+1^2} - \frac{1}{1+0^2}$$

$$= \frac{1}{2} - 1 = -\frac{1}{2}$$

Point Slope Formula: $y - y_1 = m(x - x_1)$

use $x_1 = 0$ and $y_1 = \frac{1}{1+0^2} = 1$

$$y - 1 = -\frac{1}{2}(x - 0)$$

$$y = -\frac{1}{2}x + 1$$

2. A cardiac monitor is used to measure the heart rate of a patient after surgery. It compiles the number of heartbeats after t minutes. When the data in the table are graphed, the slope of the tangent line represents the heart rate in beats per minute.

t (min)	36	38	40	42	44
Heartbeats	2530	2661	2806	2948	3080

The monitor estimates this value by calculating the slope of a secant line. Use the data to estimate the patient's heart rate after 42 minutes using the secant line between the points with the given values of t .

- (a) $t = 36$ and $t = 42$ (b) $t = 38$ and $t = 42$
 (c) $t = 40$ and $t = 42$ (d) $t = 42$ and $t = 44$

What are your conclusions?

$$A) m = \frac{2948 - 2530}{42 - 36}$$

$$m = 69.667$$

$$B) m = \frac{2948 - 2661}{42 - 38}$$

$$m = 71.75$$

$$C) m = \frac{2948 - 2806}{42 - 40}$$

$$m = 71.0$$

$$D) m = \frac{3080 - 2948}{44 - 42}$$

$$m = 66.0$$

The patient's heart rate is decreasing from 71 to 66 beats per minute after 42 minutes.